Modern Academy
for Engineering and Technology in Maadi
Electrical Engineering

Academic Year	2025-2026
Semester	FALL
Exam Date	12/11/2025
Academic Level:	Senior 2

QUESTIONS FOR THE MID TERM EXAM

Subject: (ELC 422) Digital Signal Processing **Branch:** Elec. Eng. & Comm. Tech.
Examiners: Dr. Samir M. Kamal & Dr. Sara M. Hassan **Time:** 1 hour

Number of Pages: one | Number of Questions: 2 | Attempt all questions

Question 1 [10 Points] [ILO's: a1, a2, b2, c1, c2]

a- Find the relationship between the frequencies, f, F & FS, where these frequencies respectively are: frequency of analog signals, frequency of discrete-time signals and sampling frequency. 2

b- Given the analog signal: $x(t) = 2\cos(2000\pi t)$ to be sampled with sampling frequency $F_S = 3000$Hz.

(i) Using the sampling frequency given above, could the original signal be reconstructed successfully from its samples? 1.5

(ii) Take $F_S = 8000$Hz, find an expression for its corresponding sequence, x(n). Then calculate its fundamental period N_P. 2 + 1.5

(iii) Find the samples values of the first cycle in x(n), obtained part (ii), then plot x(n). 3

Question 2 [10 Points] [ILO's: a1, a3, b2, b3, c2, c4]

a- Prove that the DFT, X(K), of a sequence x(n) is periodic. 1

b- For the following sequence: x(n) = {1, 0, 1, 0}:

(i) Using the DFT method find the frequency components of x(n) and check your results by calculating x(n) using the IFFT algorithm. 3+3

(ii) Using your results, verify Parseval's theorem. 1.5

(iii) Since we use FFT algorithm instead of DFT algorithm, what is its computational saving? 1.5

GOOD LUCK